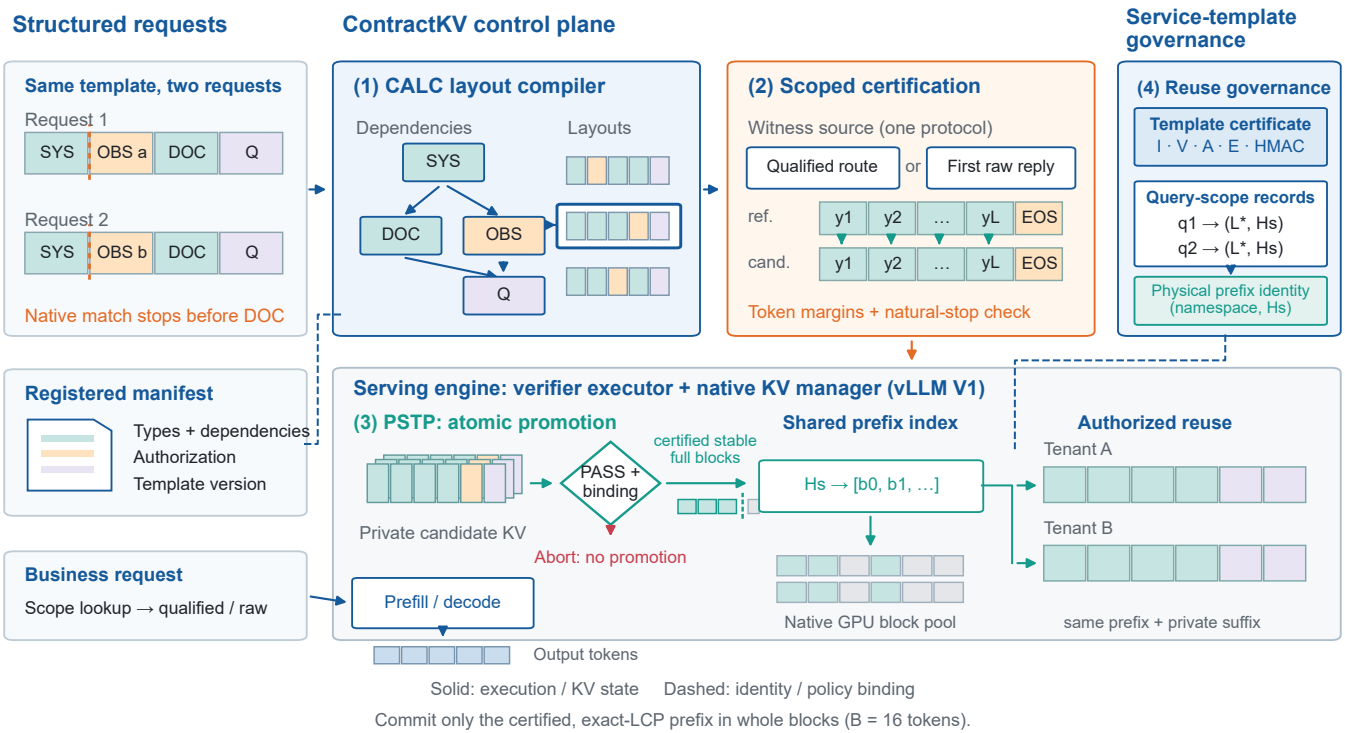
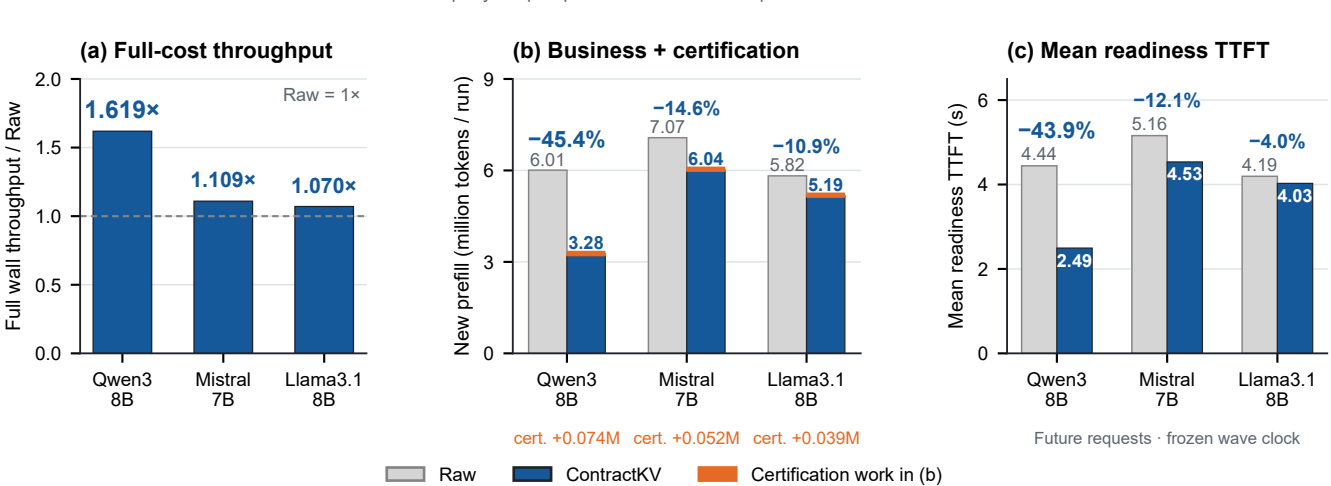
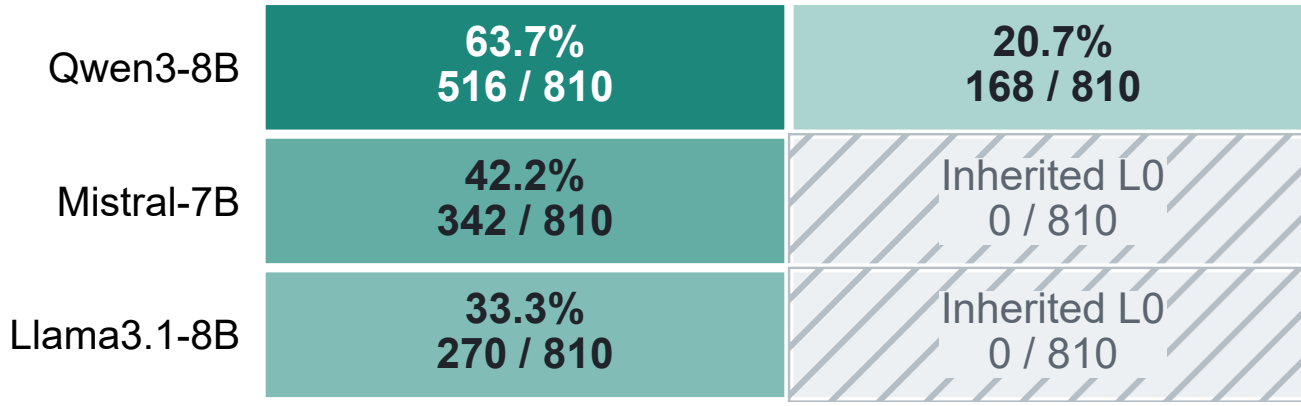

Structured requests
ContractKV control plane
Service-template governance
Same template, two requests
Request 1
SYS
OBS a
DOC
Q
Request 2
SYS
OBS b
DOC
Q
Native match stops before DOC
(1) CALC layout compiler
Dependencies
Layouts
SYS
DOC
OBS
Q
(2) Scoped certification
Witness source (one protocol)
Qualified route
or
First raw reply
ref.
cand.
y1
y2
…
yL
EOS
Token margins + natural-stop check
(4) Reuse governance
Template certificate
I · V · A · E · HMAC
Query-scope records
q1 → (L*, Hs)
q2 → (L*, Hs)
Physical prefix identity
(namespace, Hs)
Registered manifest
Types + dependencies
Authorization
Template version
Serving engine: verifier executor + native KV manager (vLLM V1)
(3) PSTP: atomic promotion
Private candidate KV
PASS + binding
certified stable full blocks
Abort: no promotion
Shared prefix index
Hs → [b0, b1, …]
Native GPU block pool
Authorized reuse
Tenant A
Tenant B
same prefix + private suffix
Business request
Scope lookup → qualified / raw
Prefill / decode
Output tokens
Solid: execution / KV state
Dashed: identity / policy binding
Commit only the certified, exact-LCP prefix in whole blocks (B = 16 tokens).


Strict query-scoped protocol · faithful 16K inputs · zero tool slack
(a) Full-cost throughput
Full wall throughput / Raw
Raw = 1×
1.619×
1.109×
1.070×
Qwen3 8B
Mistral 7B
Llama3.1 8B
(b) Business + certification
New prefill (million tokens / run)
−45.4%
6.01
3.28
−14.6%
7.07
6.04
−10.9%
5.82
5.19
cert. +0.074M
cert. +0.052M
cert. +0.039M
(c) Mean readiness TTFT
Mean readiness TTFT (s)
−43.9%
4.44
2.49
−12.1%
5.16
4.53
−4.0%
4.19
4.03
Future requests · frozen wave clock
Raw
ContractKV
Certification work in (b)

Faithful held-out · query-scoped protocol

## (a) Compiled coverage




810 future pairs / model / task family

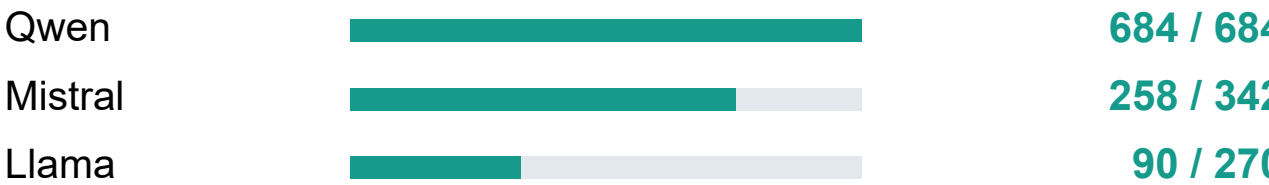
Separate RULER audit · bounded inputs · cold + fan-out



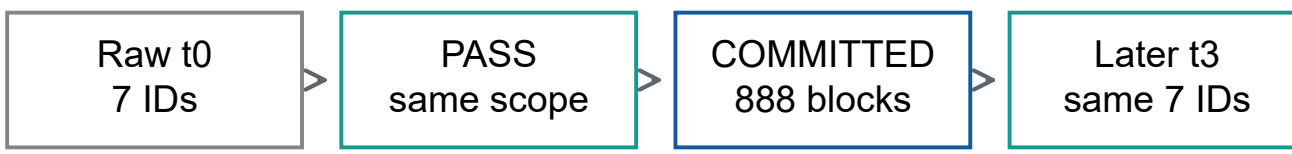

## (b) Observed output-ID differences

| | |
|---|---|
| Bounded-input (50 sources) | **0 / 2,442** |
| Faithful held-out (6 sources) | **0 / 1,296** |

Compiled subset: raw-reference score > 0

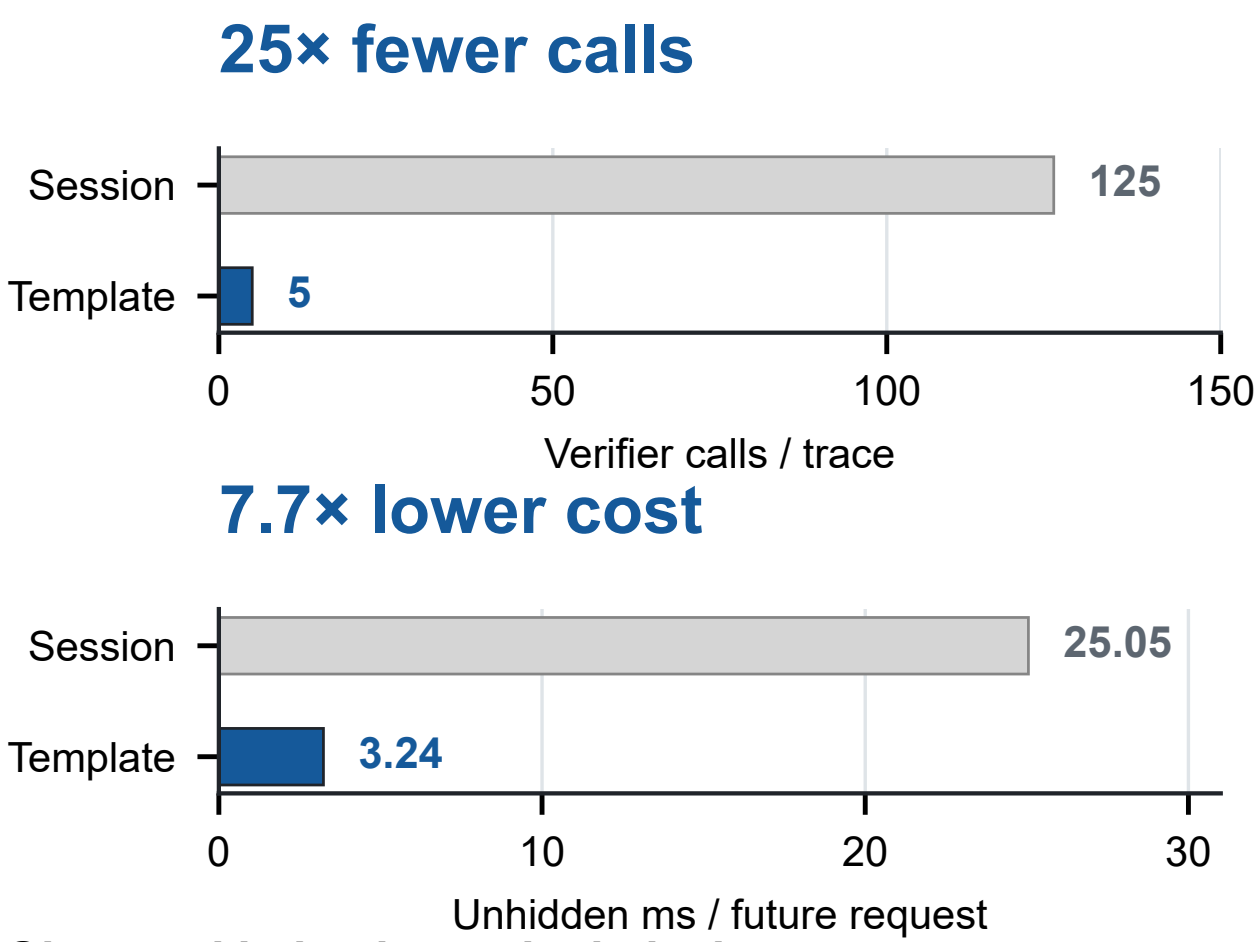



## (c) Logged Qwen / HotpotQA sequence

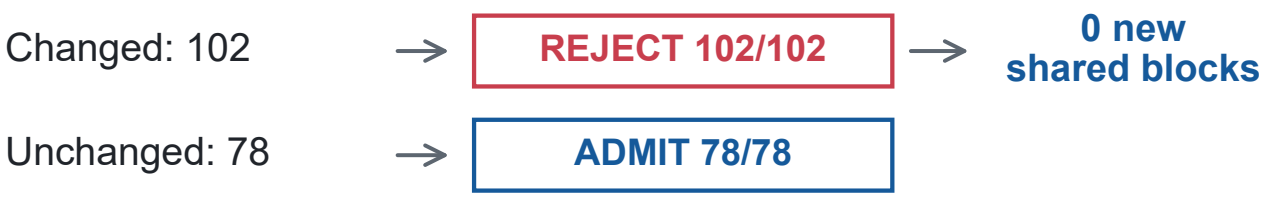



Later native hit: 14,240 tokens · records in artifact

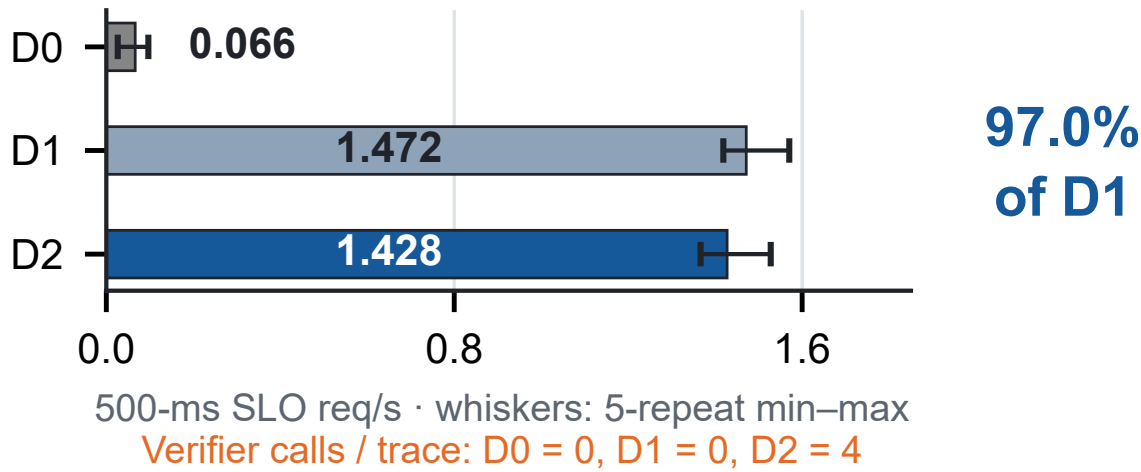
## (a) Certificate amortization



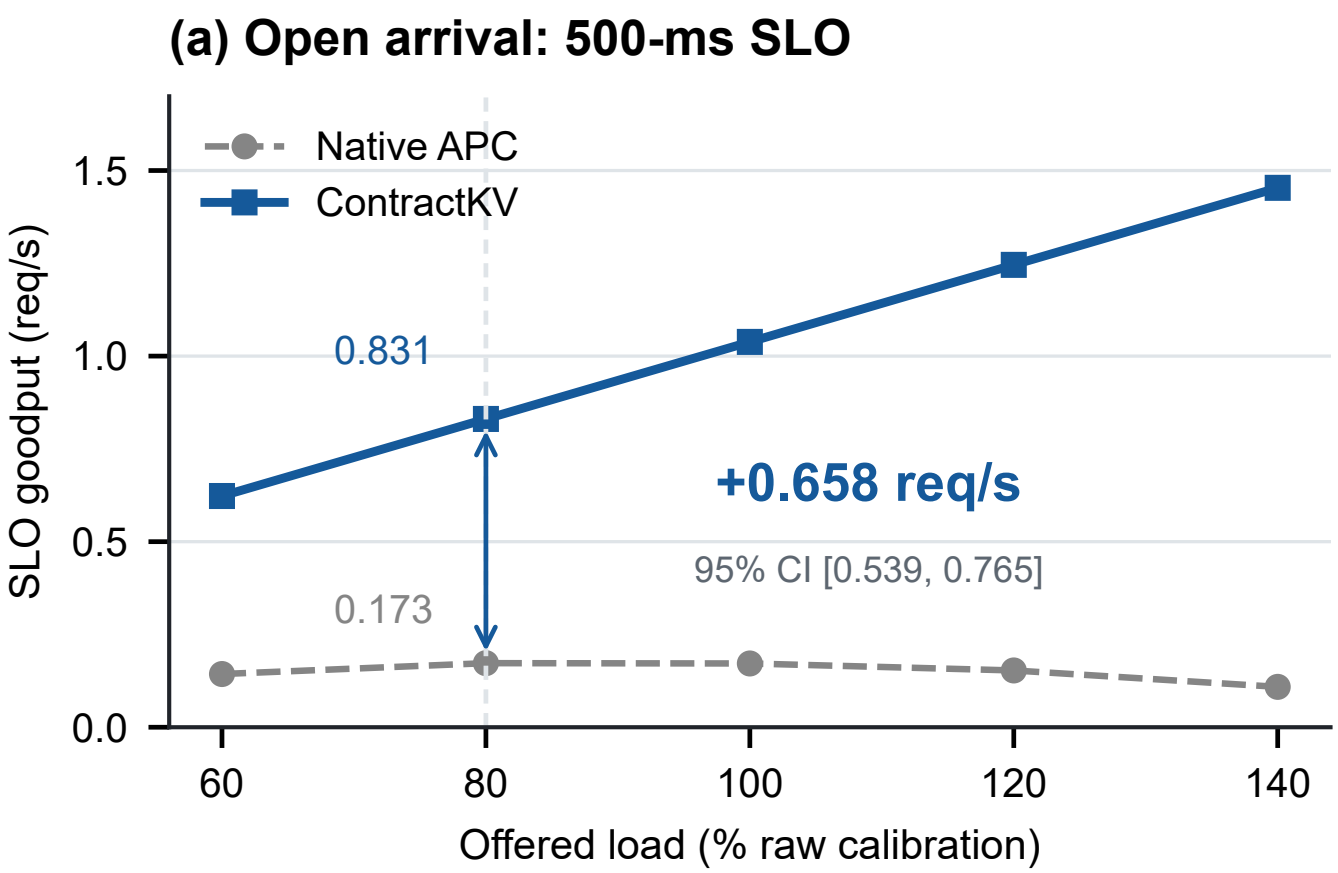

## (b) Observed behavior and admission

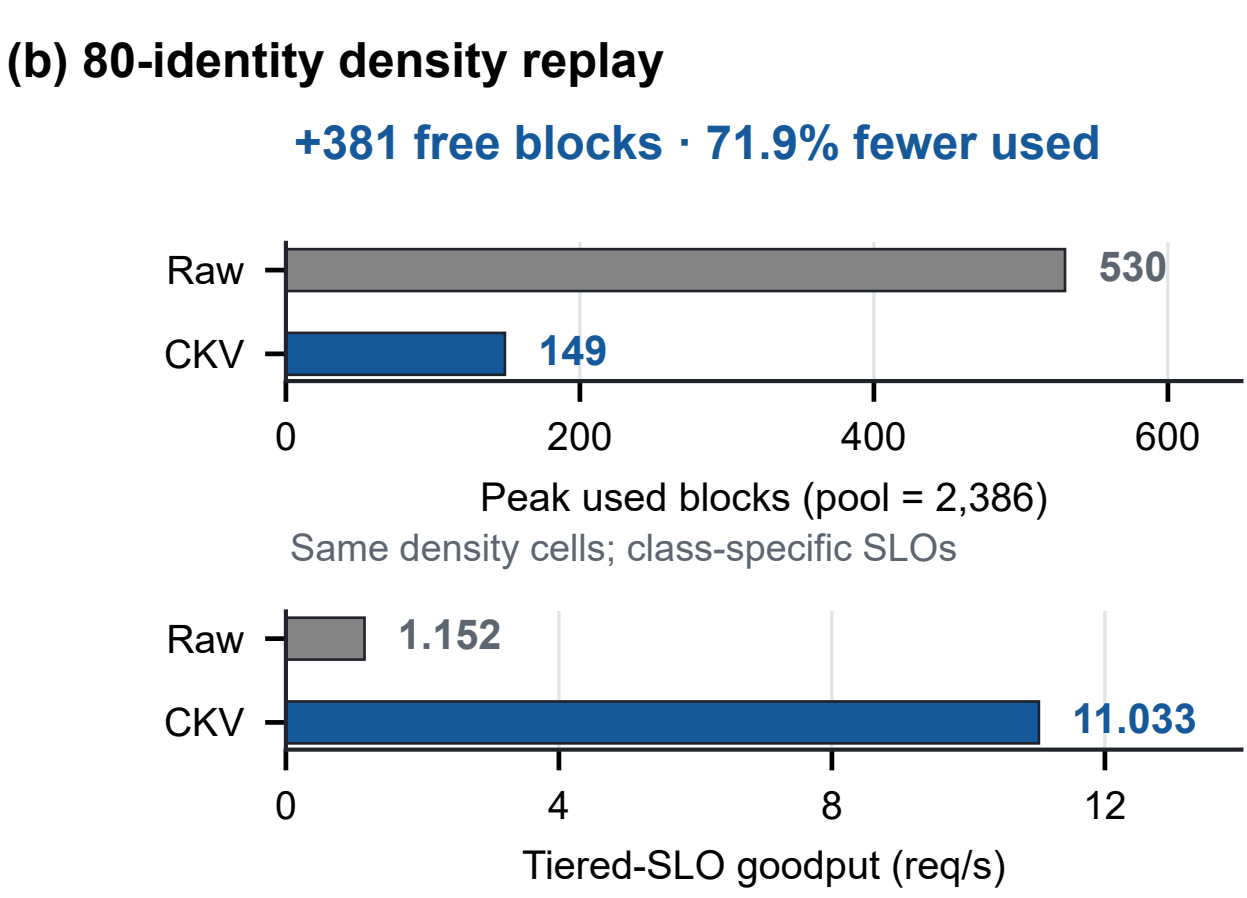



## (c) Goodput under governance



Deployment-template studies · separate protocols




Registered tenants, not concurrent GPU requests

Frozen deployment workloads

